Supplementary Reading Material: On Dual Spaces

Finite-Dimensional Vector Spaces and Dual Spaces

Definition. Let V be a vector space over a field $\mathbb{F}$. The dual space is defined as

$$V^* := \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F}).$$

Example. For $\dim V = n < \infty$, with basis $u_{i\,1\leq i\leq n}$, the dual space V^* is also n-dimensional with basis (dual basis)

$$\{f_i\}_{1\leq i\leq n}, \quad f_i : V \to \mathbb{F}, \quad u_j \mapsto \delta_{i,j}.$$

From the perspective of bilinear forms, the pairing

$$V \times V^* \to \mathbb{F}, \quad (u, f) \mapsto f(u)$$

has matrix form $I_{n\times n}$ under the bases $\{u_i\}_{1\leq i\leq n}$ and $\{f_i\}_{1\leq i\leq n}$.

Note. The isomorphism $V \simeq V^*$ is artificial. An informal but illustrative analogy: assume V has unit kilometre ($\mathbf{km}$), then V^* has unit $\mathbf{km}^{-1}$. The linear isomorphism $V \simeq V^*$ is never natural.

Example. The canonical linear map $\eta_V : V \to V^{**}$ is a natural isomorphism.

- The term "natural" signifies that for any linear map $f : U \to V$, the compositions $U \xrightarrow{f} V \xrightarrow{\eta_V} V^{**}$ and $U \xrightarrow{\eta_U} U^{**} \xrightarrow{f^{**}} V^{**}$ coincide.

Exotic Examples in Linear Algebra Without the Axiom of Choice

Example. ($\dim V = \infty$, yet $V^* = 0$). *If the axiom of choice is not assumed*, then there do exist infinite-dimensional vector spaces V with dual space $V^* = 0$. The example by H. Läuchli presents an infinite-dimensional vector space V whose proper subspaces are all finite-dimensional.

Example. (When V^* is strictly larger than V, yet $V \simeq V^{**}$). *In ZF set theory with dependent choice*, it takes effort to verify $\mathbb{R}[[x]]^* \simeq \mathbb{R}[x]$. However, $\mathbb{R}[[x]]$ does not possess a countable basis.

Proposition. The canonical linear map $V \to V^{**}$ is injective if and only if V is a subspace of some product space $\prod_X \mathbb{F}$. For instance, $\mathbb{F}[x]$ and $\mathbb{F}[[x]]$ both embed into their double dual spaces.

Proof. See this article.

Axiom of Extension

Axiom. (Axiom of Extension, also known as the Hahn-Banach Axiom (HB)).

- For any linear subspace $U \subseteq V$, every linear map $f : U \to W$ extends to a map $\widetilde{f} : V \to W$ such that $\widetilde{f}|_U = f$.

Remark. The Hahn-Banach Axiom is independent of the usual axioms (ZF + DC), which are assumed in the study of tensor products of infinite-dimensional vector spaces.

Exercise. The Hahn-Banach Axiom is equivalent to the assertion that $\mathrm{Hom}_{\mathbb{F}}(-, W)$ always maps injections to surjections.

Exercise. By HB, for any $v_1, v_2 \in V$ with $v_1 \neq v_2$, there exists $f \in V^*$ such that $f(v_1) \neq f(v_2)$. Hence, the canonical linear map $V \to V^{**}$ is always injective.

Exercise. By HB, show that for $U \hookrightarrow V$, the following map is an isomorphism

$$\frac{V^{**}}{U^{**}} \to (V/U)^{**}, \quad [x]_{U^{**}} \mapsto \pi^{**}(x),$$

where π is the natural quotient map $V \twoheadrightarrow V/U$.

Dual Spaces and Galois Correspondence

Definition. (Kernels and Annulateurs). For convenience, let U_i denote subspaces of V, and B_i subspaces of V^*.

- $\mathrm{ann}\, U := f \in V^* \mid f(U) = 0$, a subspace of V^*;
- $\ker B := u \in V \mid f(u) = 0\ (\forall f \in B) = \bigcap_{f \in B} \ker f$, a subspace of V.

These notions are defined for subsets in general. By the closure property:

- $\mathrm{ann}(X) = \mathrm{ann}(\mathrm{span}(X))$, for $X \subseteq V$;
- $\ker(S) = \ker(\mathrm{span}(S))$, for $S \subseteq V^*$.

Remark. $\ker f$ and $\ker f$ are identical.

Exercise. Show that for any finite-dimensional vector space V, the map $\mathrm{ann} : \text{Subspaces of } V \to \text{Subspaces of } V^*$ is a bijection with inverse map $\ker$.

Proposition. One has $\mathrm{ann}(\ker B) \supseteq B$. Equality does not generally hold.

Proof. The formula $\mathrm{ann}(\ker B) \supseteq B$ involves logic more than algebra. For a counterexample, take V as the space of continuous functions over $\mathbb{R}$, and $B = \mathrm{span}(\{\delta_r\}_{r \in \mathbb{Q}})$.

Exercise. (Galois Connection)

1. $\mathrm{ann}(\ker B) \supseteq B$,

2. $\ker(\operatorname{ann} U) \supseteq U$,
3. $\operatorname{ann}(\ker(\operatorname{ann} B)) = \operatorname{ann}(B)$,
4. $\ker(\operatorname{ann}(\ker B)) = \ker B$.

The pair $(\ker, \operatorname{ann})$ defines a Galois correspondence.

Example. For a set map $f : X \to Y$, the image and preimage maps (f_*, f^*) form a Galois connection between $\mathrm{Subset}(X)$ and $\mathrm{Subset}(Y)$.

Definition. Let $Op(\mathbb{R}^2)$ denote the set of open subsets of $\mathbb{R}^2$, and $Cl(\mathbb{R}^2)$ the set of closed subsets. Define:

1. $k : Op(\mathbb{R}^2) \to Cl(\mathbb{R}^2), \quad U \mapsto \overline{U}$ (closure);
2. $i : Cl(\mathbb{R}^2) \to Op(\mathbb{R}^2), \quad K \mapsto K^\circ$ (interior).

Exercise. Show that $ikik = ik$ and $kiki = ki$; hence, (k, i) form a Galois connection.

Exercise. Given $X \subseteq \mathbb{R}^2$, show that at most seven distinct sets (including X) can be obtained by applying k and i repeatedly.

Exercise. Show that one can obtain at most 14 distinct sets (including X itself) by applying i and $(-)^c$ repeatedly.

The Yoga of Dual Spaces

Theorem. In the following table, FD denotes finite-dimensional cases, G denotes general cases, and HB refers to cases where the Hahn-Banach Axiom is assumed.

For a linear map $\varphi : U \to V$, define the dual map as

$$\varphi^* : V^* \to U^*, \quad [V \to \mathbb{F},\ v \mapsto f(v)] \mapsto [U \to \mathbb{F},\ u \mapsto f(\varphi(u))].$$

No.	Formula	FD	G	HB
G1	$\operatorname{ann}(\ker B) \overset{?}{=} B$	$=$	$\supseteq$	$\supseteq$
G2	$\ker(\operatorname{ann} V) \overset{?}{=} V$	$=$	$\supset$	$=$
E1	$\operatorname{ann}(V_1 + V_2) \overset{?}{=} \operatorname{ann}(V_1) \cap \operatorname{ann}(V_2)$	$=$	$=$	$=$
E2	$\operatorname{ann}(V_1 \cap V_2) \overset{?}{=} \operatorname{ann}(V_1) + \operatorname{ann}(V_2)$	$=$	$\supseteq$	$=$

No.	Formula	FD	G	HB
E3	$\ker(B_1 + B_2) \overset{?}{=} \ker B_1 \cap \ker B_2$	$=$	$=$	$=$
E4	$\ker(B_1 \cap B_2) = \ker B_1 + \ker B_2$	$=$	$\supseteq$	$\supseteq$
Z1	$\ker \varphi^* \overset{?}{=} \operatorname{ann}(\operatorname{im} \varphi)$	$=$	$=$	$=$
Z2	$\ker \varphi \overset{?}{=} \ker(\operatorname{im} \varphi^*)$	$=$	$\subseteq$	$=$
Z3	$\operatorname{im} \varphi^* \overset{?}{=} \operatorname{ann}(\ker \varphi)$	$=$	$\subseteq$	$=$
Z3'	$\operatorname{im} \pi^* \overset{?}{=} \operatorname{ann}(\ker \pi)$, with π surj.	$=$	$=$	$=$
Z4	$\operatorname{im} \varphi = \ker(\ker \varphi^*)$	$=$	$\subseteq$	$=$
Z5	If f is surj., then f^* is inj.	true	true	true
Z6	If f is inj., then f^* is surj.	true	NaN	true
L1	$S^* \overset{?}{\to} V^* / \operatorname{ann}(S)$	$\simeq$	$\hookrightarrow$	$\simeq$
L2	$(V/S)^* \overset{?}{\to} \operatorname{ann}(S)$	$\simeq$	$\simeq$	$\simeq$

Remark. When the general case (G) coincides with the HB case, the proposition is provable without additional axioms. When the FD case differs from the G case, the FD part is proved using dimension arguments. Nonexamples disprove the HB part of statements G1 and E4.

- G1-G2 come from Galois connection,
- E1-E4 are elementary,
- Z1-Z6 are taken from the textbook of 张贤科,
- L1-L2 are taken from the textbook of 黎景辉等.